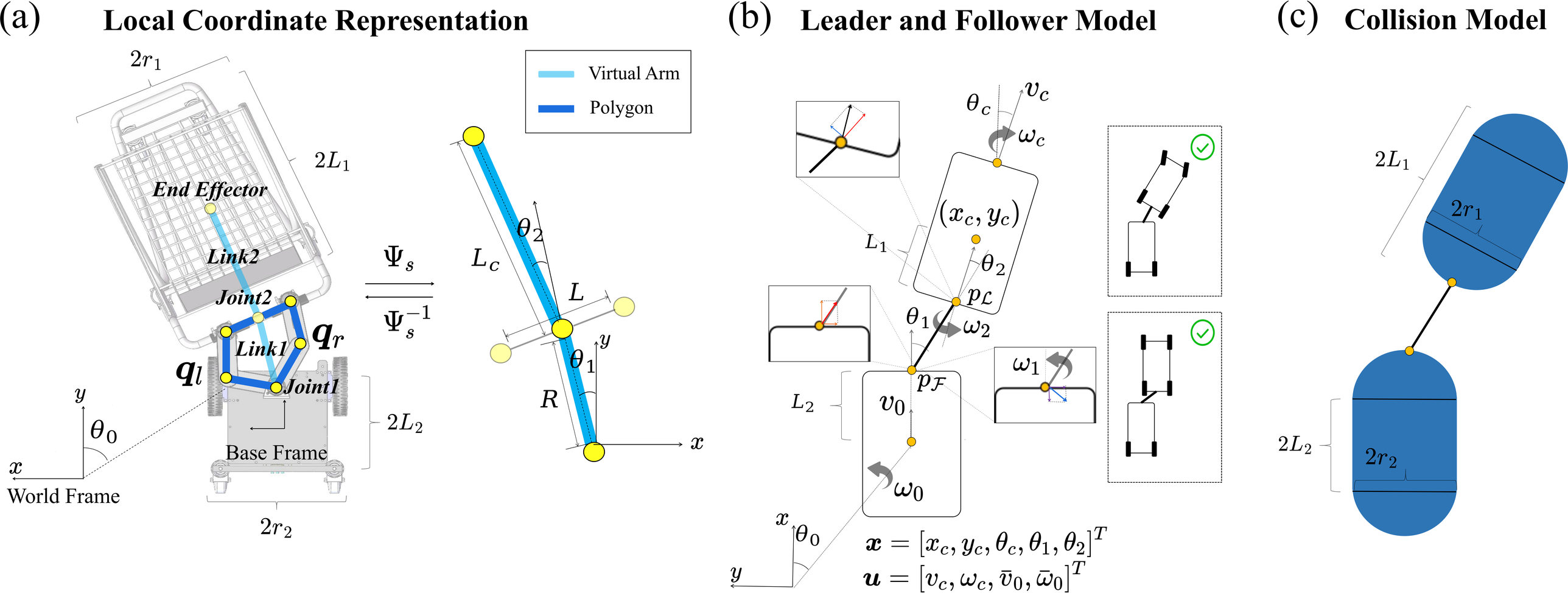

(a)
Local Coordinate Representation
$2r_1$
$2L_1$
End Effector
Link2
Joint2
Link1
Joint1
q_r
q_l
$2L_2$
Base Frame
$2r_2$
y
θ_0
x
World Frame
Ψ_s
Ψ_s^{-1}
Virtual Arm
Polygon
θ_2
L_c
L
y
θ_1
R
x
(b)
Leader and Follower Model
θ_c
v_c
ω_c
(x_c, y_c)
L_1
θ_2
$p_{\mathcal{L}}$
θ_1
ω_2
$p_{\mathcal{F}}$
ω_1
L_2
v_0
ω_0
x
θ_0
y
$\boldsymbol{x} = [x_c, y_c, \theta_c, \theta_1, \theta_2]^T$
$\boldsymbol{u} = [v_c, \omega_c, \bar{v}_0, \bar{\omega}_0]^T$
(c)
Collision Model
$2L_1$
$2r_1$
$2L_2$
$2r_2$